407 Electromagnetic Induction

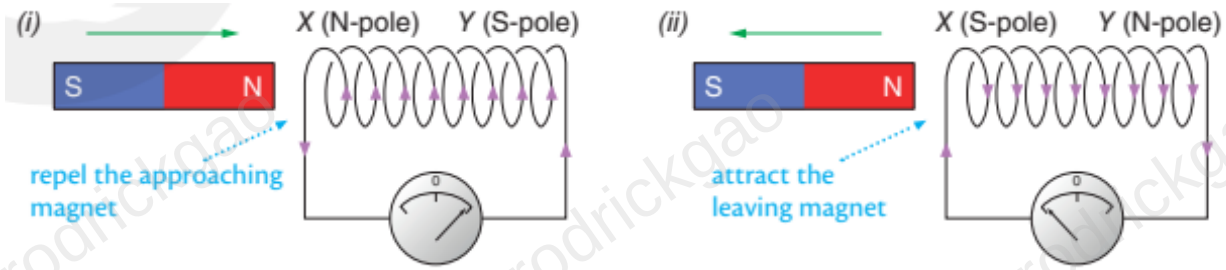
4.7.1 · 4.7.2 · 4.7.3 Electromagnetic induction

4.7.1 · 4.7.2 Conditions for electromagnetic induction

Changing magnetic flux linkage induces an e.m.f.; faster change gives a greater magnitude. Changes in field, coil area or orientation can cause induction, so mechanical motion is not essential. An induced e.m.f. drives a sustained current only when a closed conducting path is present.

4.7.3 Lenz's law

The magnetic effect of induced current opposes the change causing it, not always the original field itself. If original flux increases, the induced field opposes the increase; if it decreases, the induced field supports the original direction to oppose the decrease. Identify the flux change first, then use the right-hand rule to determine current direction.

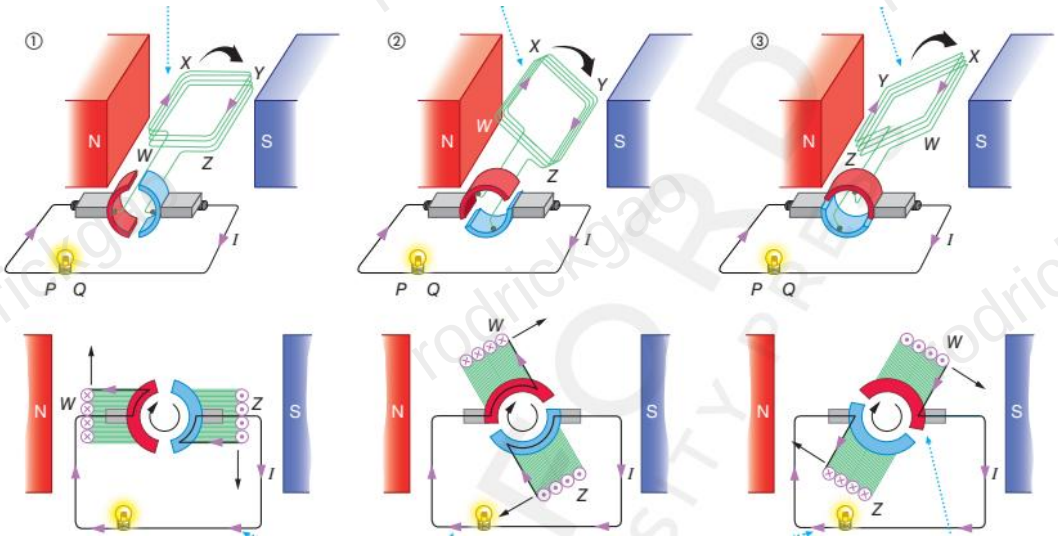


The induced field opposes the flux change caused by an approaching or receding magnet.

Textbook sources: 407 · PDF 5, 6

4.7.4 A.c. and d.c. generators

Rotating a coil in a magnetic field changes its flux linkage periodically and generates alternating e.m.f. Slip rings maintain each coil end's external connection, producing a.c. A split ring exchanges connections every half-turn, producing unidirectional but pulsating external current. Generation converts mechanical energy into electrical energy.



Rotation changes flux linkage; the commutator makes the external current unidirectional but pulsating.

Textbook sources: 407 · PDF 16, 18

4.7.5 Eddy currents

Changing flux through conducting loops within a bulk conductor induces circulating eddy currents. Their magnetic effects oppose the change and can provide magnetic braking; their resistive heating is used in induction heating. Laminating transformer cores with insulated thin sheets restricts large circulating paths and reduces these losses.

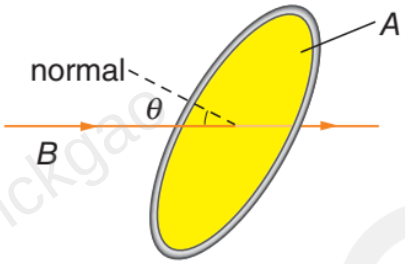


Fig 7.3c *B making an angle θ with the normal of the coil.*

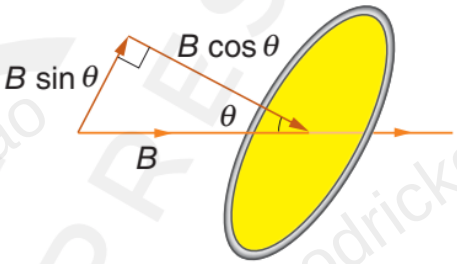


Fig 7.3d *Components of B parallel and perpendicular to the normal of the coil.*

θ is the angle between B and the area normal. Only B cos θ passes normally through the area; flux is zero when the field is parallel to the surface.

Textbook sources: 407 · PDF 22, 31

4.7.6 · 4.7.8 Magnetic flux and area orientation

Magnetic flux through a flat surface in a uniform field equals flux density times the area projected perpendicular to the field. Measure the angle from the surface normal: flux is largest when the field is

perpendicular to the plane and zero when parallel. Flux, measured in Wb, differs from flux density, measured in T.

$$\Phi = BA\cos\theta$$

$$1\ \mathrm{T} = 1\ \mathrm{Wb\ m^{-2}}$$

Uniform field through a flat area: Φ (Wb); B (T); A (m²); θ: angle between B and the surface NORMAL, not the plane.

Magnetic flux density equals flux per perpendicular area for a uniform field.

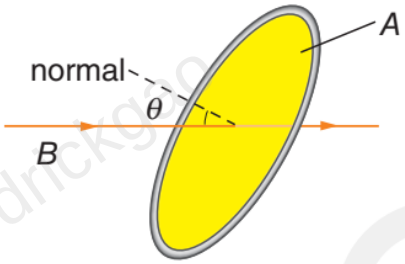


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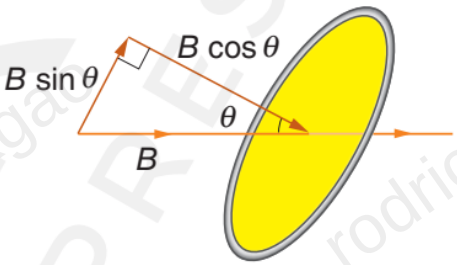


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Textbook sources: 407 · PDF 31

4.7.7 · 4.7.9 · 4.7.10 Flux linkage and induction

4.7.7 Magnetic flux linkage

Flux linkage is the sum of the flux linked by each turn. It equals number of turns times flux per turn only when every turn links the same flux. Increasing turns increases the induced e.m.f. for the same per-turn flux change. Do not confuse one turn's flux with the linkage of the entire coil.

$$\Psi = N\Phi$$

Ψ: magnetic flux linkage (Wb turn); N: turns; Φ: flux through each turn (Wb). All turns link the same flux.

4.7.9 · 4.7.10 Faraday's law

Induced e.m.f. magnitude equals the rate of change of flux linkage. Average e.m.f. uses the linkage change divided by elapsed time; instantaneous e.m.f. follows the tangent slope of a flux-linkage-time graph. Take the absolute value when only magnitude is required. For signed values, choose consistent normal and winding directions; the minus sign expresses Lenz's law.

$$\varepsilon_{\text{av}} = -\frac{\Delta\Phi}{\Delta t}(N = 1)$$

$$\varepsilon_{\text{av}} = -N\frac{\Delta\Phi}{\Delta t}$$

ε: average induced e.m.f. (V); Φ: flux per turn (Wb); N: fixed turns; Δt: s. Minus sign expresses Lenz's law. Instantaneous e.m.f. uses the instantaneous rate of change.

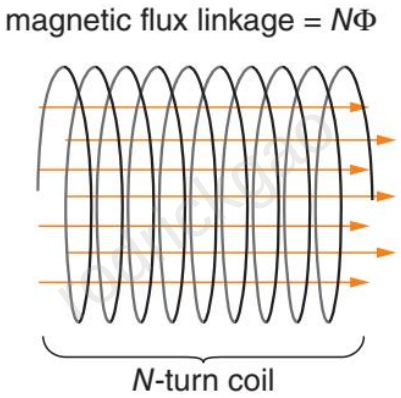
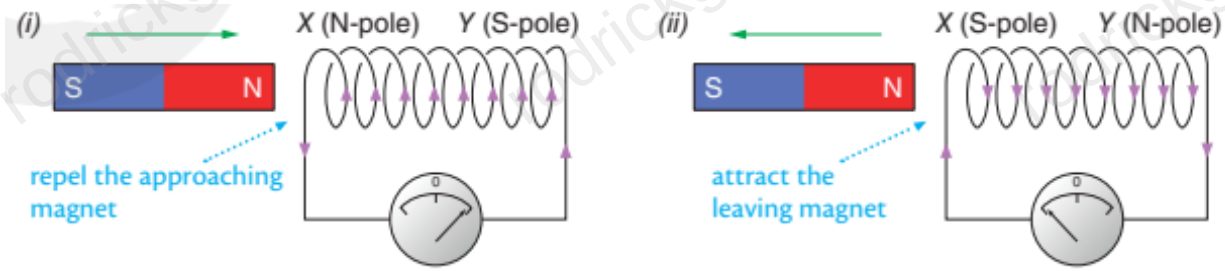


Fig 7.3e *Magnetic flux linkage through an N-turn coil.*

Each of N turns links flux Φ, so total flux linkage is NΦ. A change in this total determines the induced e.m.f.



The induced field opposes the flux change caused by an approaching or receding magnet.